

XDL Generator

“Move bottle to tray,
transfer the contents of beaker into flask0,
and then Shake flask for 30s.”



Finetuned Llama 3.2 1B

XDL (Chemical Description Language)

<Procedure>

<Move object=“bottle” place=“tray” />

<Transfer from_vessel=“beaker” to_vessel=“flask” />

<Stir vessel=“flask” time=“30” />

</Procedure>

XDL Validator

Syntactic Check

<Move object=“bottle” place=“tray” />

<Transfer from_vessel=“beaker” to_vessel=“flask” />

<Shake vessel=“flask” time=“30” />

Object-Existence Check

<Move object=“tube” place=“tray” />

<Transfer from_vessel=“beaker” to_vessel=“flask0” />

<Stir vessel=“flask0” time=“30” />

Physical Feasibility Check

<Move object=“bottle” place=“tray0” />

<Stir vessel=“flask” time=“30” />

Usable XDL Tag

Move

Stir

Transfer

CleanVessel

HeatChill

Add

Shake

World State



tube



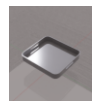
beaker



flask



bottle



tray

Invalid XDL

Valid XDL

TAMP Module